

S2.5 Case Study - Developing a new FM strategy for 610.000m2 real estate

MARTIJN DE RIET

BIMFORCE



bimforce



GR CENTRE

LJUBLJANA

11 – 13 OCT 2018



Our team is built around knowledge and experience. With over 30 years of combined work on BIM projects, we know the challenge.



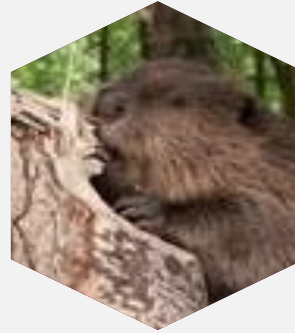
MARTIJN DE RIET
OWNER

Information Specialist,
focussed on developing
solutions



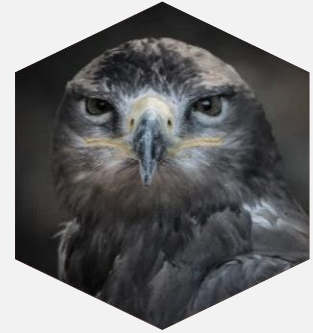
MARK WIERINGA
OWNER

Information Specialist,
allround IFC guru



ROEL RUIJTERS
LEAD DEVELOPER

If we can imagine it, Roel
can build it



OWEN SLOOTWEG
PROJECT MANAGER

Keeping a sharp eye on
our projects, planning and
budgets



Goals

Our goal is to assist our clients with creating their Information Infrastructure.

We research client workflows, information needs and current bottlenecks
And implement our Information Management Solutions to fit custom client needs



Some numbers for the TUD.



The Technical University of Delft is one of the oldest and most respected universities in the Netherlands. Some numbers:

CAMPUS

Founded in 1842
1,600,000 m² campus
610,000 m² build environment

PEOPLE

21,000 students
6,500 staff size
25 technostartups
200 companies on campus

BUILDINGS

88 buildings on campus
62 buildings owned by TUD
Ranging from nuclear
research facilities to student
housing

BUILDING INFORMATION

1 projects designed/ constructed
using BIM (more in 3D / hybrid)
300m paper archive
48.000 digital documents

Electronic Building Dossier

A BRIEF HISTORY OF TIME

Use case

DIGITAL AGE

More and more processes and information is digitalized. M&O organisations need to follow.

It's becoming increasingly difficult for M&O to interact with other processes and outside companies in an analog way.

END USER DEMANDS

Building occupants no longer accept that it takes weeks to fix something or go through a paper archive to find something.

LAWS & REGULATIONS

Building owners face more and more backlash if things go wrong, don't work or take too much time to fix.

Regulations are moving to shift accountability to owner. Users are more aware of this

But most importantly

TIME

It takes up a lot of time for building owners to find their stuff. If they ever do...

And if you do find it, there is no guarantee that the information is correct.

FLEXIBILITY

Building owners are in it for the long run. They have one certainty:

The tools they use today will **not** be the tools they use in 30 years from now.

RELIABILITY

Building data is stored in tens of different applications, files, documents, spreadsheets, and so on.

An answer in one file can contradict the answer in another file.

Project requirements

The EBD-project was founded on some basic project requirements

FAST & RELIABLE

Proven technology, not necessarily from AEC
Single source of truth

SAFE & TRUSTING

Open Source solutions
Should provide TU Delft with guaranteed access to information

PLUG & PLAY

Changes in software shouldn't have effect on primary processes
Building Information is the basis, software is a tool
Centralized information storage

FUTURE PROOF

New developments, latest technology
Requires Open Standard data structure

CUSTOMER OWNERSHIP

TU Delft owns data
TU Delft should be able to host on premise

EFFICIENT

Should decrease search times for information requests
Should decrease maintenance time on drawings, CAD and models

Project setup

The EBD-project was meant to be developed and used simultaneously

PILOT PROJECT: TNW ZUID LABORATORY

30.000m² of laboratory

INTEGRATED INFORMATION

“Full BIM” project required.

With enough challenges to simulate real-world application



Project setup

The EBD-project was setup with a collaborative approach

R&D PROGRAM

Focussing on developing new solutions. This entails the possibility of failure.

OPEN COLLABORATION

Between TU Delft, Maintenance consortium and Bimforce

SCRUM

Using Scrum as workflow for the Planning stage:

- Bi-weekly progress meetings
- Trello as scrumboard

LESSONS LEARNT:

- What do all parties want to take away from the project? Besides the obvious project goals
- Scrum is not ideal for execution phase (coding progress is not interesting for building owner)



DEFINITION OF INFORMATION FOR A BUILDING OWNER

What is Information?

The definition of Information can vary greatly, depending on your role in the AECO process.

According to Wikipedia:

Information, in its most restricted technical sense, is a sequence of symbols that can be interpreted as a message.

Information is any kind of event that affects the state of a dynamic system that can interpret the information.

What is Information?

These definitions share one single attribute:

*Information, in its most restricted technical sense, is a sequence of symbols that can be **interpreted** as a message.*

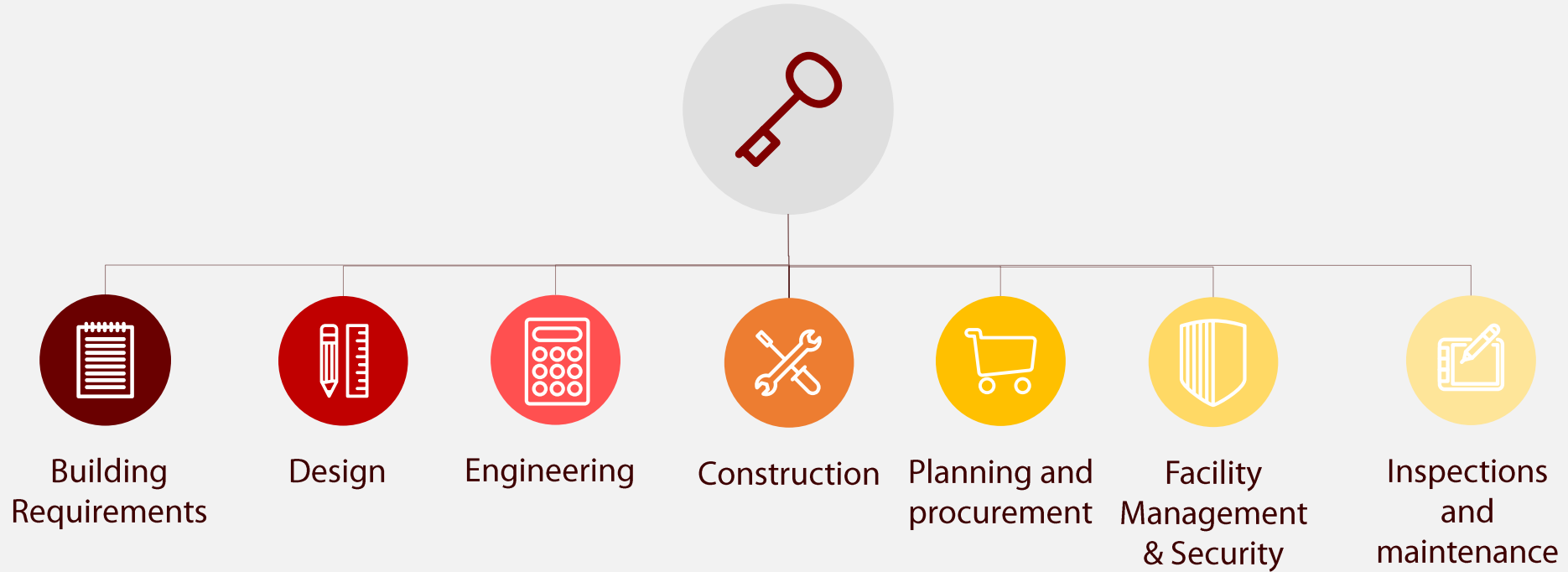
*Information is any kind of event that affects the state of a dynamic system that can **interpret** the information.*

What is Information?



SENDING STUFF
TO BE INTERPRETED

Key aspects of Building Information



Once upon a time...

One key difference between AEC and OM:

Time.

Time means a transition from a
linear process to a networked process

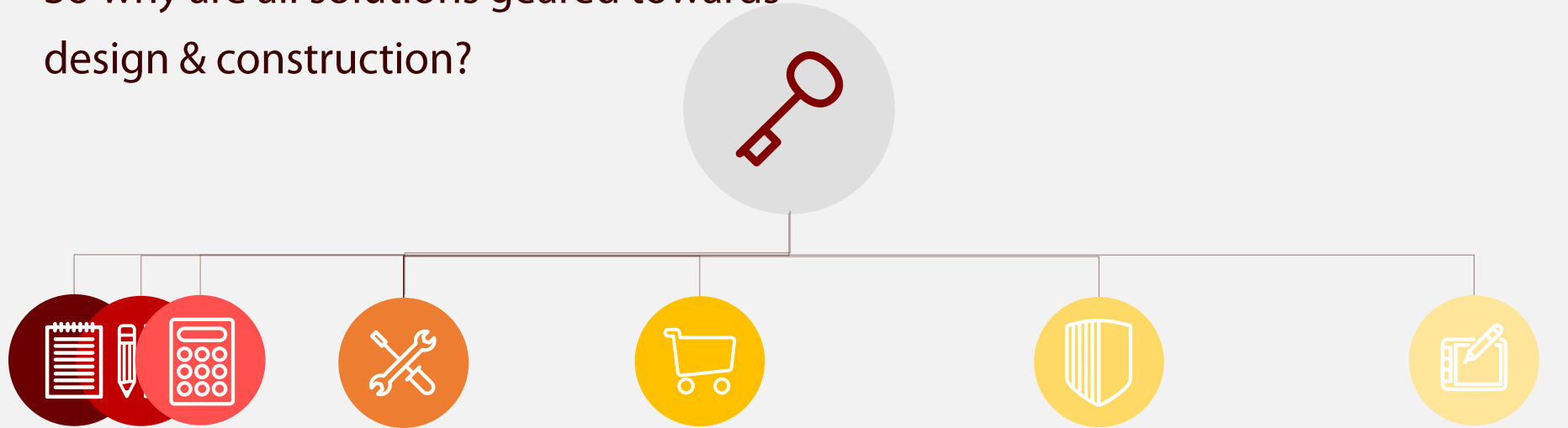
Time means that a static representation
(a file) by definition does not suffice



Relative importance

Time means that relative costs become a leading aspect.

So why are all solutions geared towards design & construction?



Project requirements

Focus of the EBD **solely** on Operation and Maintenance. Which means:

1. Software independent
(and that includes IFC!)
2. Based on Open Standards
(not necessarily IFC!)
3. Data driven
(no files, not even IFC)
4. Full data-ownership and data-governance by TUDelft
(right to transfer, right to be forgotten)

Database requirements

1. Scalable
2. Dynamic data structure
3. Relations between entities should be queryable



Privacy & Security

1. User Management should comply with GDPR standards
2. Building data should be treated similar to Users



Privacy & Security

Data protection by design:

- No User information stored in GRiDS
- No user recognizable in GRiDS

Security by Design:

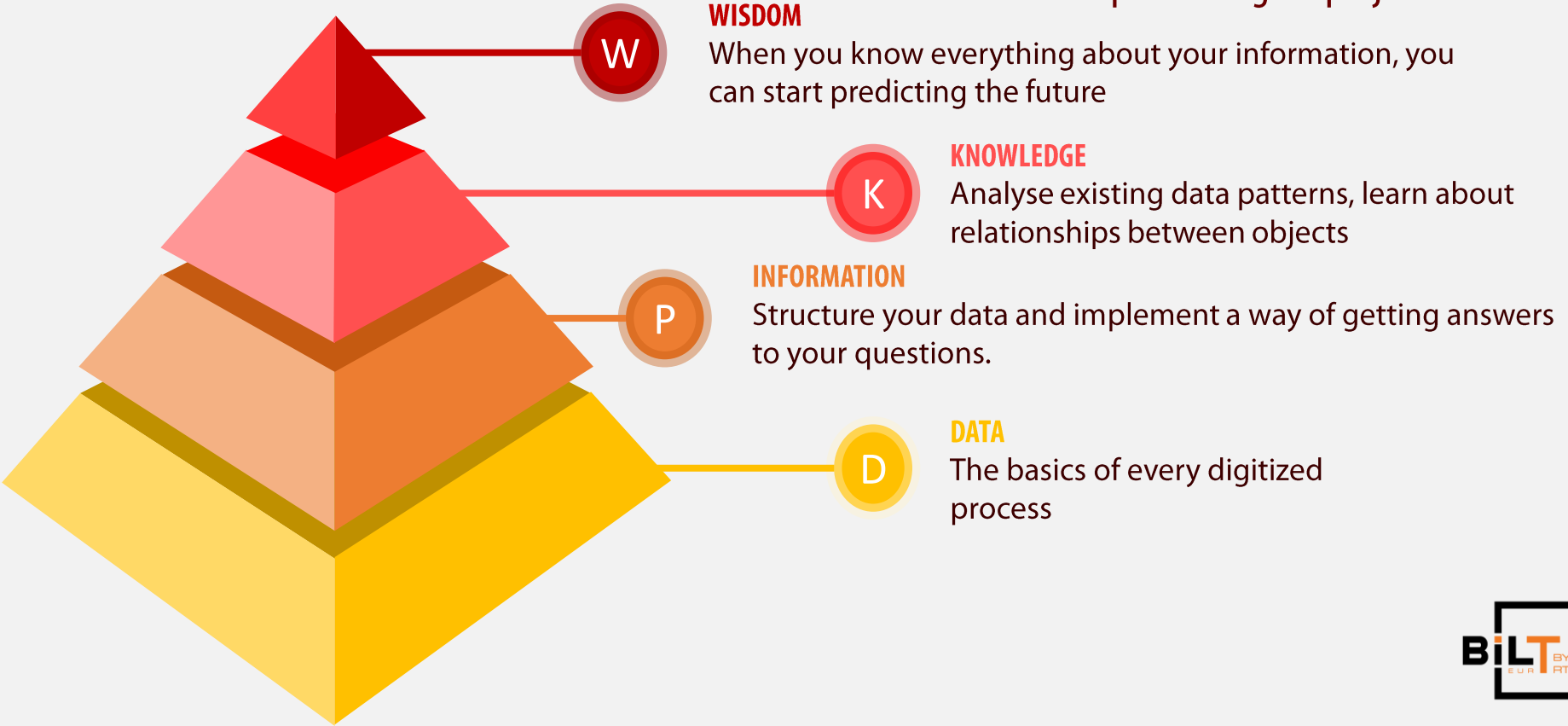
- Using an API for all communications
- On premise hosting



RESEARCH: BASIC CONCEPT FOR THE ELECTRONIC BUILDING DOSSIER

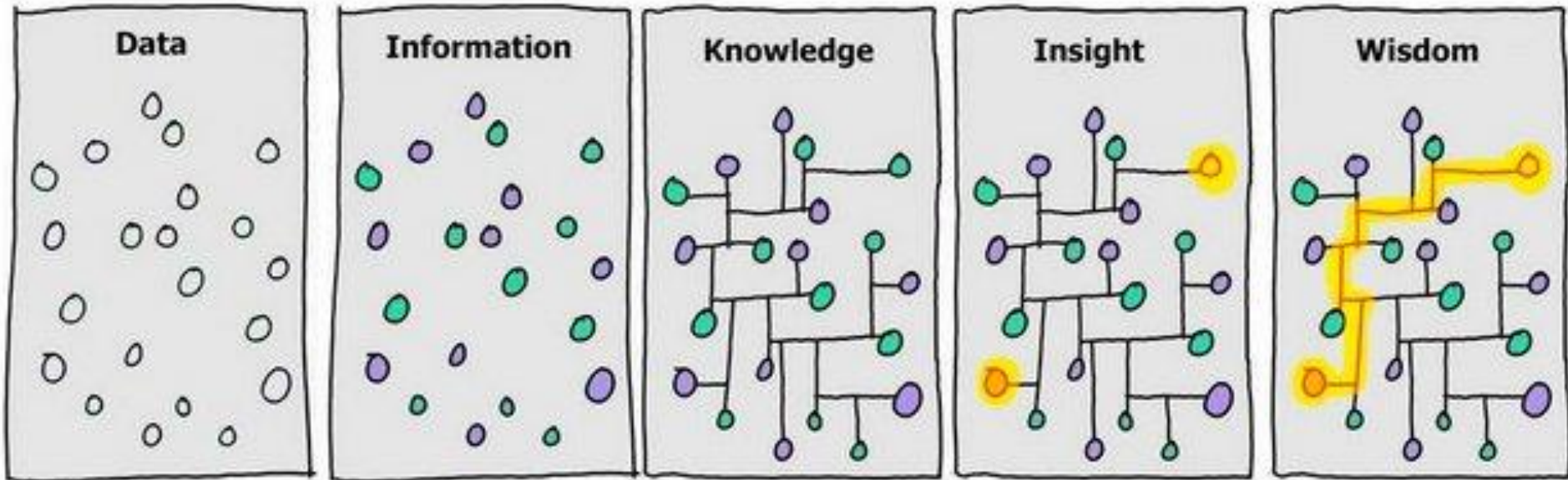
Evolution of Data

In a perfect digital project



Evolution of Data

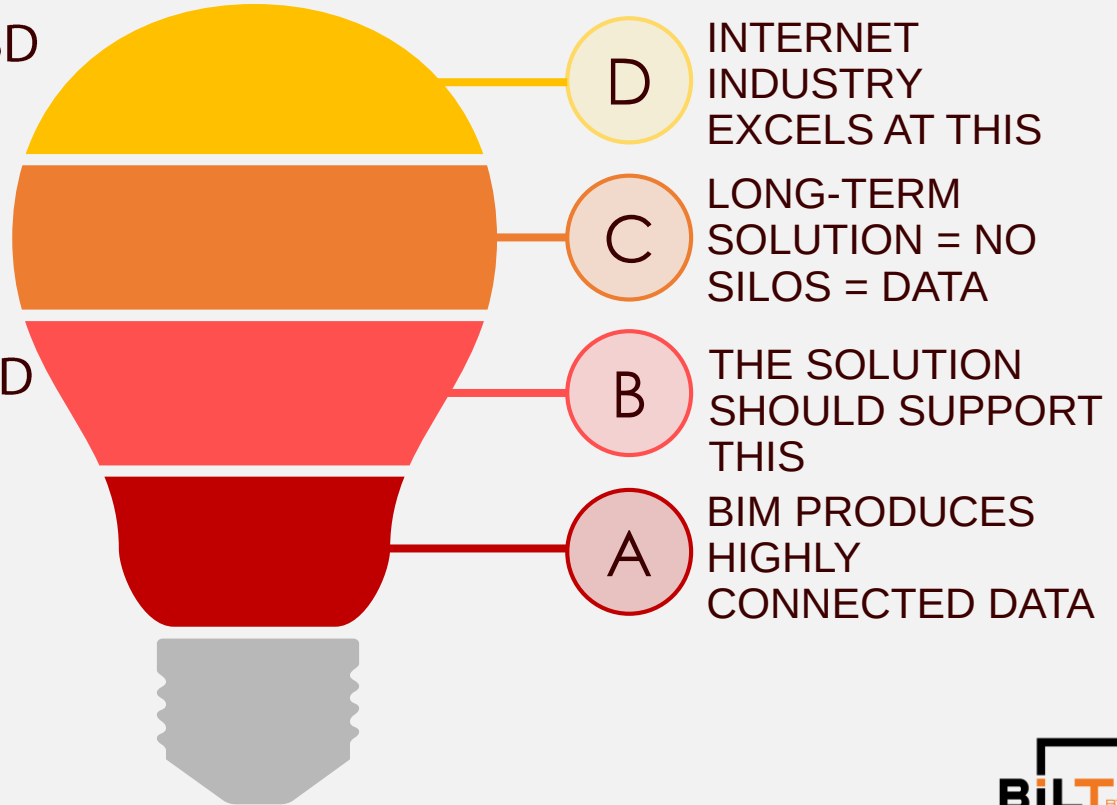
In a perfect digital project



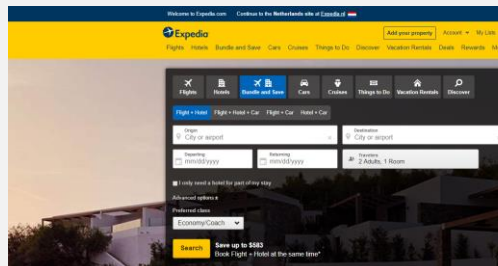
Look outside!

The central principle of the EBD was to create a solution that

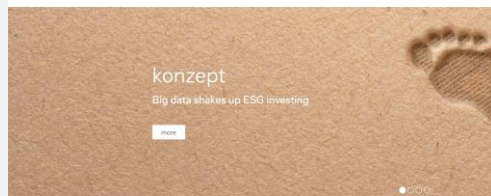
- Would comply to the requirements set by the TUD
- Would use existing technologies from the one field that specializes in “big data”



Where to look?



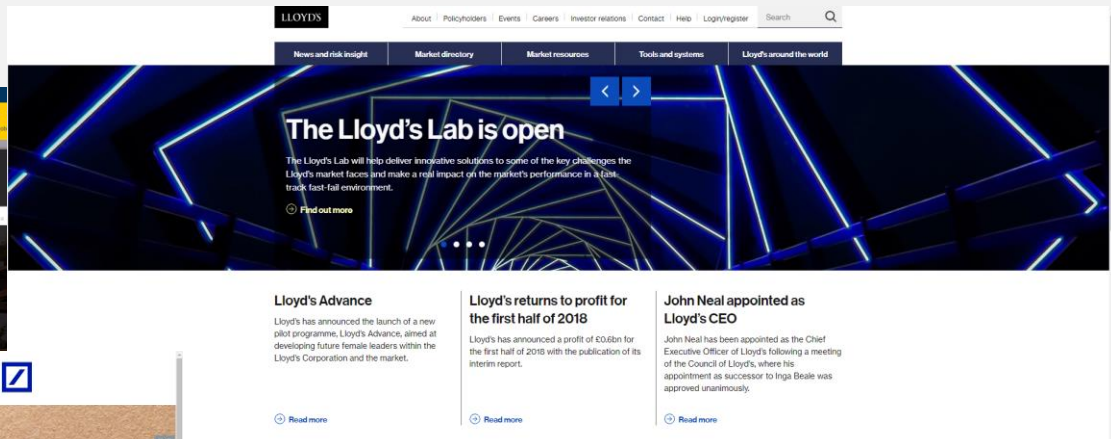
Deutsche Bank Company News Careers Investor Relations Research Responsibility #HustleImpact
Client login Products Location finder



You are here: Home



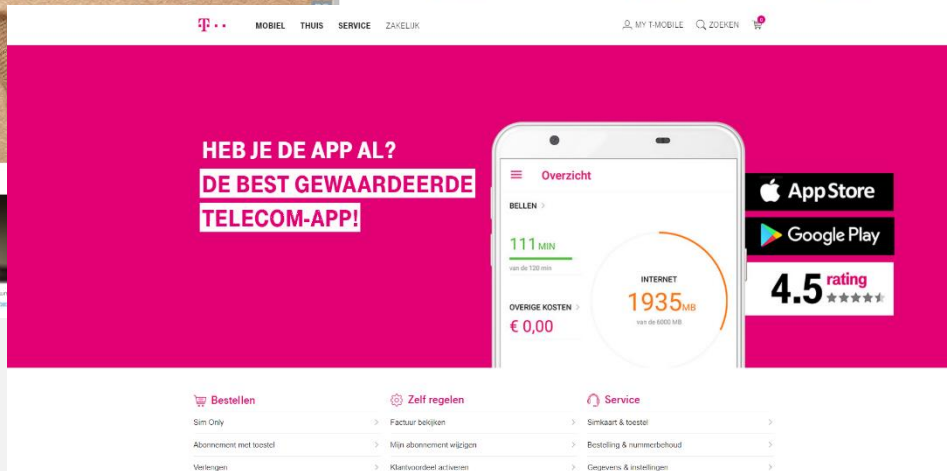
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Read more

Read more

Read more



Bestellen

Sim Only

Abonnement met toestel

Verlengen

Zelf regelen

Factuur betalen

Mijn abonnement wijzigen

Klantvriendelijke acties

Service

Simkaart & toestel

Bestelling & nummerbehoud

Gegevens & instellingen

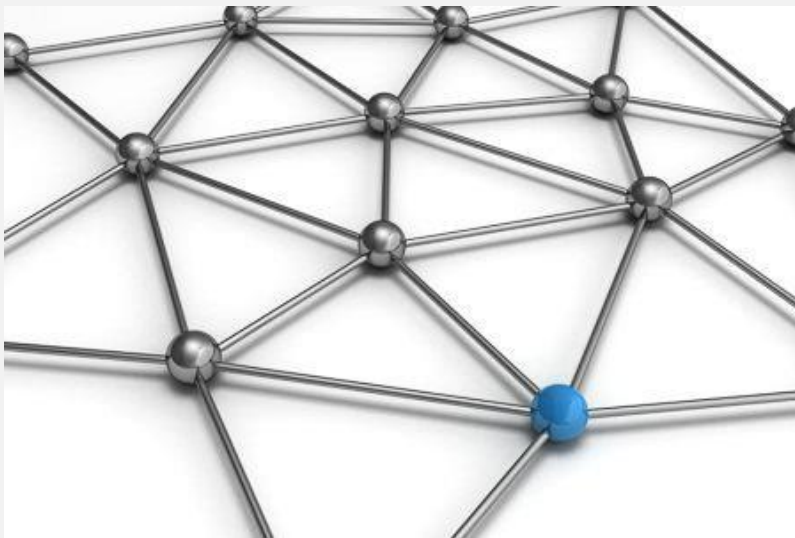


Where to look?

The answers are all around us. We use them every day.

Linked data systems across the web:

1. Booking industry;
2. Online shopping;
3. Insurance & risk management;
4. Online banking & payments;
5. Telecom providers;



What's the challenge?

Big data handles relatively small datastructures in large numbers.

Shopping:

Buyer, articles, search history, recommendation engines

Fraud detection:

Person, transaction, receiving party, sending party

Datasets can grow to billions of occurrences, which are analysed using webbased algorithms.

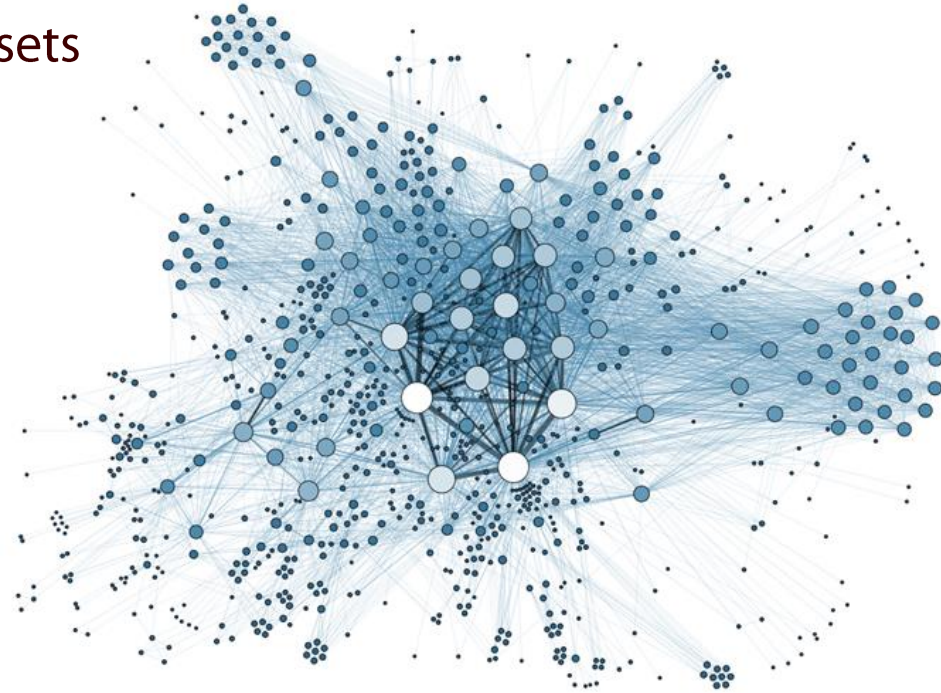
What's the challenge?

Buildings consist of relatively small datasets
(maybe up to 1 million occurrences?)

Which are infinitely more complex...

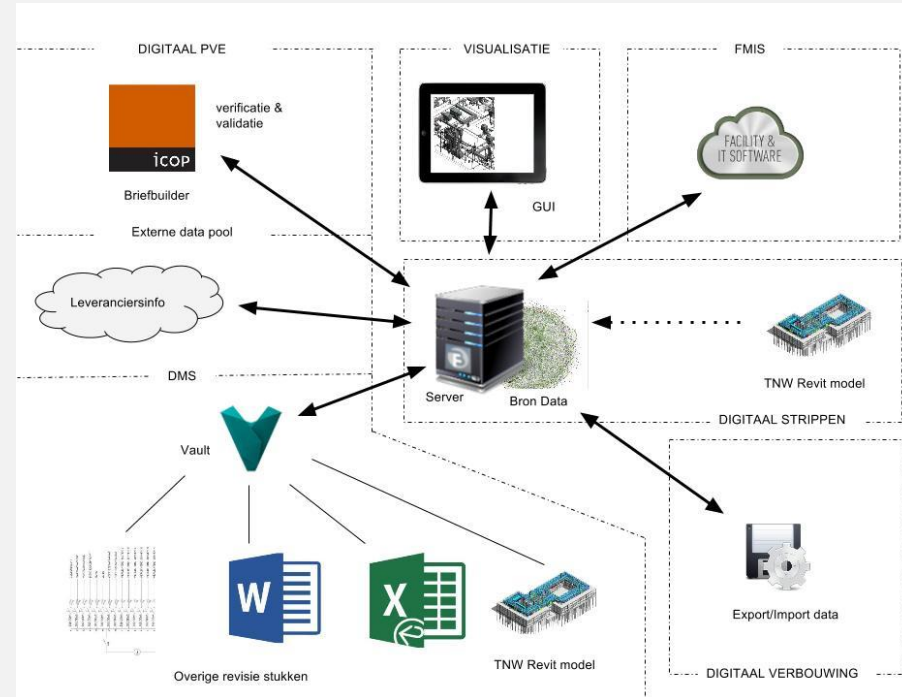
Input methods (software you want to
connect) isn't standardized or stable

Information requirements change over
time.



Functional concept

- Digital takeoff
- Data Visualisation
- Automated requirement verification
- Document management system
- External data pool
- Export/Import API
- Preliminary steps in linking FMIS



Technical concept

Or as we called it:

Project Deathstar.

The ultimate weapon in
Information Management



And yes, the first few iterations did kind of blow up

But like the Empire, we don't know when to stop

Technical concept

DEATHSTAR PROJECT SCHEMA

bimforce 1 October 13, 2016
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Interface Layer:

This is where you operate the Deathstar

Application Layer:

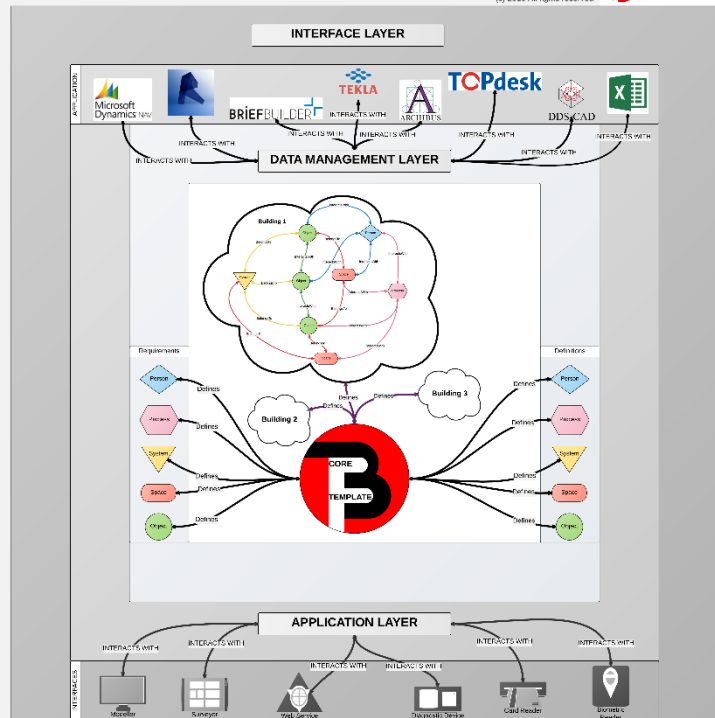
Communication with your tools.

Data Management Layer:

Handling and unifying data structures

Core database:

IFC based single source of truth



LET'S GO TO WORK



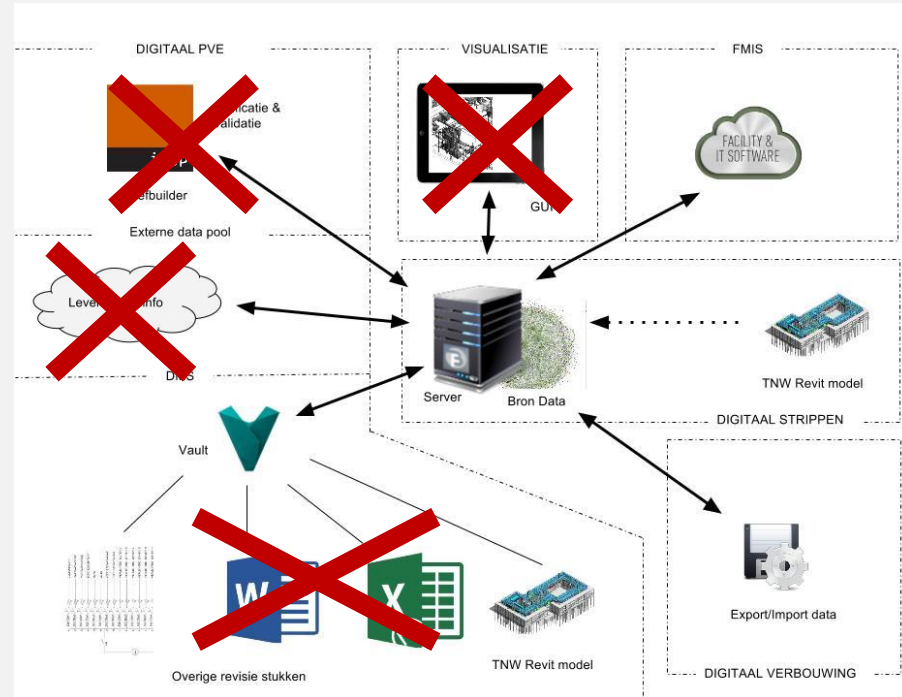
The first bump(s)

- Project funding exceeded EU regulations for public procurement.
- Scope proved to be too extensive.
- The technical concept was flawed (more on that later)



How we solved them

- Scope got reduced.
- Which meant no public procurement.
- But also meant we had to rethink the technical concept
- Because we were still going to do the rest, but as separate projects.

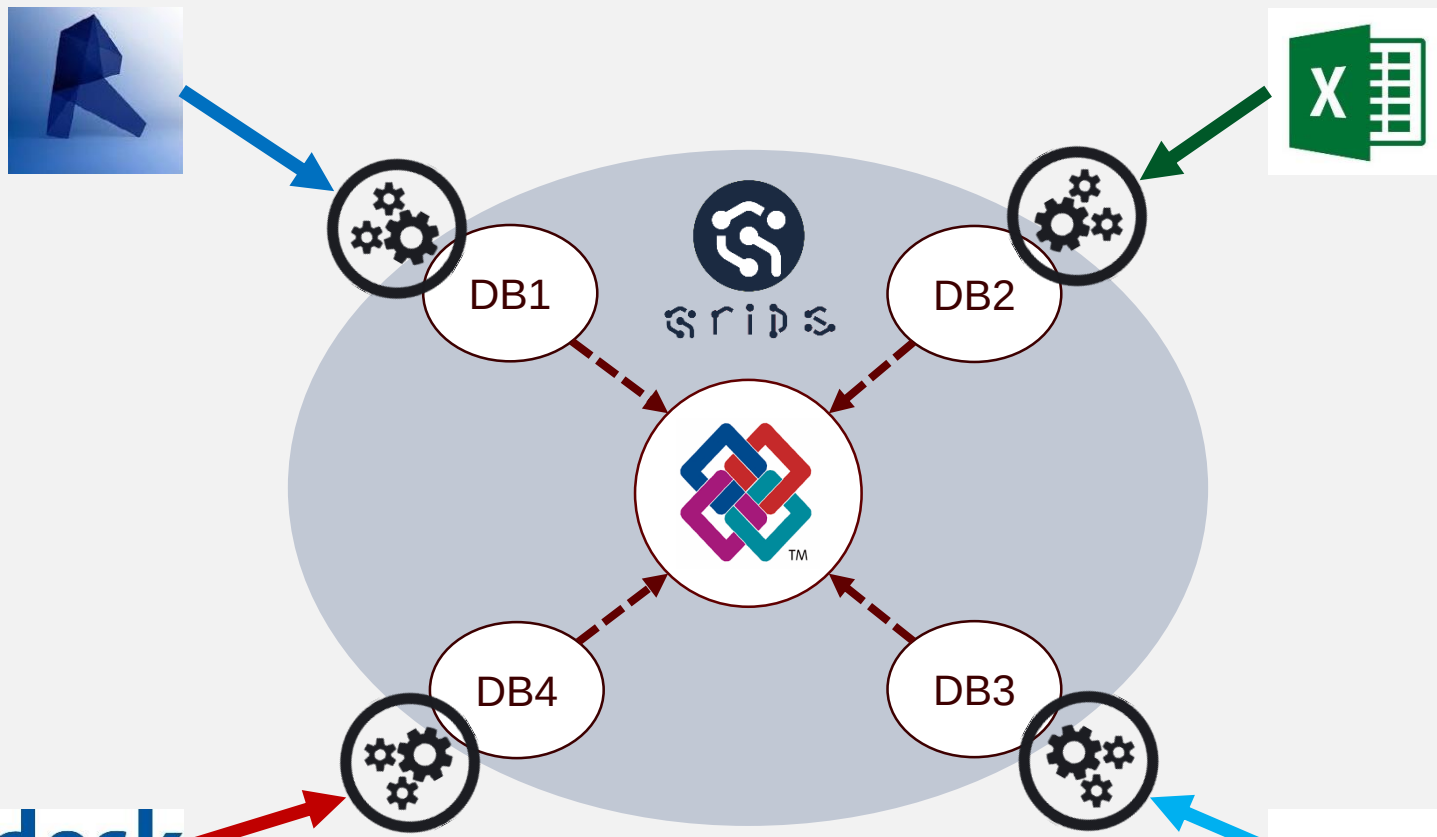


Second technical iteration

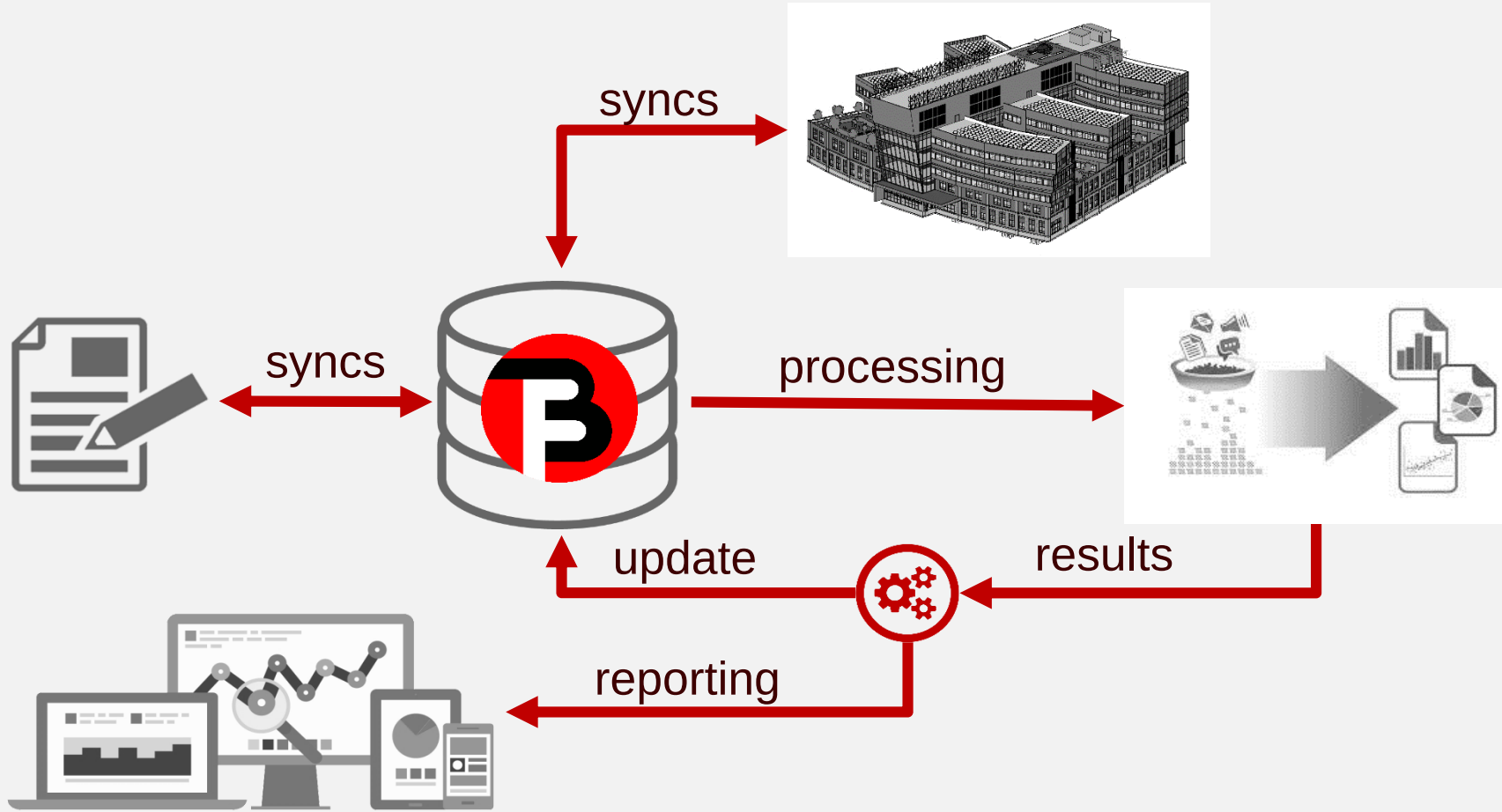
- Introduction of a two-layered database
- Each application has it's own sub-database, structured according to application data
- Central IFC database combines application-specific data (if needed)
- Communication types:
file | plugin | webservice



Second technical iteration



Second technical iteration



First Alpha version – Project Overview



[Instructions](#) [Contact](#)

Project Overview

Number	Name	Active?	Action
BIM Engineer-NJ17-Groep1	GRIDS pilot project voor Groep 1, Avans+ BIM Engineer NJ17	True	SetInactive
BIM Engineer-NJ17-Groep2	GRIDS pilot project voor Groep 2, Avans+ BIM Engineer NJ17	True	SetInactive
ere	ere	True	SetInactive
GRIDS_dev	Development project GRIDSv1.0	True	SetInactive
RRU_Test	TET RRU	True	SetInactive
SRON_SO	SO verification	True	SetInactive
SRON_VO+	V+O verification	True	SetInactive
SRONTest2	Test for SRON Briefbuilder Link	True	SetInactive
Test_Rolin	Ultprobeersel	True	SetInactive
Test_TIB	gebouw	True	SetInactive
UMCU Test2	Testfile UMC Utrecht	True	SetInactive

Add Project

Object Linker

Query Builder

Project Details

Export Project

First Alpha version – Object Linker



[Instructions](#) [Contact](#)

Object Linker for

DocumentSet	GROEP1-TO
Source	-- Select List (0) --
DocumentItem	-- Select List (7) --
ObjectType	-- Select List (292) --



DocumentSet	GROEP1-TO
Source	-- Select List (0) --
DocumentItem	-- Select List (1) --
ObjectType	-- Select List (2) --

Property	Operator	Value
Area	>	1000
Area	=	
Area	=	

Show Result

Property	Operator	Value
-- Select property (32) --	=	
-- Select property (32) --	=	
-- Select property (32) --	=	

Show Result

result (75)

- Algemeen sanitair familie 03.225
- bedieningsruimte MRI 03.220
- berging (technische dienst) 03.055
- berging 03.221
- berging 03.275
- berging/schoon 03.035
- Dagverblijf sport 03.010
- Dagverblijf huiskamer 03.130
- Dagverblijf huiskamer 03.135
- elekra/data 03.105
- elekra/data 03.205
- elekra/data 03.235

result (67)

- 1. Kantoren, vergaderruimten en archief
 - 1.1 Kantoren medewerkers
 - 1.1.01 Directie
 - 1.1.02 Hooft
 - 1.1.03 Sectiehoofd/projectleiders
 - 1.1.04 Overige medewerkers en ondersteunend
 - 1.1.05 Gecombineerd werkplekken/labs (opgenomen in 4.2, 5.1 en 6.4)
 - 1.1.06 Afstudeerders/stagaires
 - 1.1.07 Bezoekers/tijdelijke werkplekken
 - 1.2 Vergaderruimten
 - 1.2.01 Grote vergaderzaal (2 zalen, om te vormen tot 1 zaal, 75 pers.)
 - 1.2.02 Vergaderzaal tot 20 personen

[Project Overview](#) [Query Builder](#) [Project Details](#) [Export Project](#)

First Alpha version – Query Builder



[Instructions](#) [Contact](#)

Query Builder

SRON - Revit Masses totaal inclusief Design Option
SRON - Revit Masses Design Option '00_PVE varianten.variant 3'
SRON - Overzicht van alle Ruimten en RuimteGroepen
SRON-Overzicht van alle Briefbuilder ruimten en de Revit ruimten die hieraan gekoppeld zijn
SRON-Overzicht van alle Briefbuilder ruimten die niet gekoppeld zijn aan Revit Elementen
SRON-Overzicht van alle Revit Rooms die niet gekoppeld zijn aan Briefbuilder Ruimten
SRON-Controle minimum aantal ruimten Briefbuilder met aanwezige ruimten in Revit model
SRON-Controle minimaal oppervlak van ruimten Briefbuilder met daadwerkelijke oppervlak ruimten in Revit model

Show entries

Search:

RuimteNummer	RuimteGroep	GrossFloorArea	DesignOptionID
1.1	Kantoren, vergaderruimten en archief	824,0022000000007	00_PVE varianten : blokkenschema vereenvoudigd
1.1	Kantoren, vergaderruimten en archief	92,88000000000011	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	58,32000000000006	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	213,8400000000002	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	168,4800000000002	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	56,16000000000007	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	14,39999999999994	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	178,2000000000003	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	165,2400000000001	00_PVE varianten : variant 1
1.1	Kantoren, vergaderruimten en archief	120,9599999999991	00_PVE varianten : variant 2

Showing 1 to 10 of 211 entries

[Previous](#) [1](#) [2](#) [3](#) [4](#) [5](#) ... [22](#) [Next](#)

[Download as Excelfile](#)

[Replace Queryfile](#)

[Project Overview](#)

[Query Builder](#)

[Project Details](#)

[Export Project](#)



Third technical iteration - beta

- Introduction of an intermediate hybrid datastructure
- Upgrade of the User Interface
- Implementation of connectors:
 - File import / export
 - Plugin Revit
 - WebAPI for webservice



IMPLEMENTATION AND TESTING

TESTING

- First datasets are being published
- Let's take a look
(with a demonstration model)

GRiDS platform

- Aimed at harvesting & integrating data from software tools
- Allows users to validate information, analyse and report the results
- Provides the required Information Management structure to re-use existing data



Project Overview



[Help](#) [Contact](#)



Project Setup > Import > Object Linker > Query Builder > Verification > **Information Takeoff**

Project:	Choose Project
Document Set:	Choose DocumentSet
Source:	Select a DocumentSet first
DocumentInfo:	Select a DocumentSet first

Object Category:	Select a DocumentSet first
Object Property:	Select a Object Category first
Property Value:	Select a Object Property first

No Data, complete above selections first

Show In Viewer

Import Briefbuilder

[Help](#)[Contact](#)[Project Setup](#)[Import](#)[Object Linker](#)[Query Builder](#)[Verification](#)[Information Takeoff](#)

Project:	Choose Project
Document Set:	Choose DocumentSet
Source:	Select a DocumentSet first
DocumentInfo:	Select a DocumentSet first

Object Category:	Select a DocumentSet first
Object Property:	Select a Object Category first
Property Value:	Select a Object Property first

No Data, complete above selections first

[Show In Viewer](#)

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Object Linker



[Help](#) [Contact](#)



Project Setup > Import > **Object Linker** > Query Builder > Verification > Information Takeoff

DocumentSet	-- Select List (0) --
Source	-- Select List (0) --
Document	-- Select List (0) --
ObjectCategory	-- Select List (0) --

Property	Operator	Value
-- Select property (0) --		
-- Select property (0) --		
-- Select property (0) --		

Show Result

DocumentSet	-- Select List (0) --
Source	-- Select List (0) --
Document	-- Select List (0) --
ObjectCategory	-- Select List (0) --

Property	Operator	Value
-- Select property (0) --		
-- Select property (0) --		
-- Select property (0) --		

Show Result

Project Overview | Query Builder | Project Details | Export Project

Query Builder

[Help](#)[Contact](#)

Project Setup Import Object Linker **Query Builder** Verification Information Takeoff

Available Queries

VO+ Verificatie SRON

Overzicht van alle Briefbuilder ruimten en de Revit ruimten die hieraan gekoppeld zijn

Overzicht van alle Briefbuilder Ruimten die niet gekoppeld zijn aan Revit Rooms

Overzicht van alle Revit Rooms die niet gekoppeld zijn aan Briefbuilder Ruimten

Controle aantal aanwezige ruimtes

5 10 20 50 100 250

Page 1 of 2 (8 items) 1 2



BriefbuilderDocument	BriefbuilderRuimte	BriefbuilderURL
Q	Q	Q
SRON-Leiden	Uit algemeen PvE	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	Buitenruimte	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	Parkeerplaats auto's	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	6.6. Parkeerplaatsen	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	Rijwielberging inpandig (180 fietsen + 8 motoren)	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	6.5. Rijwielberging inpandig (180 fietsen + 8 motoren)	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	Schoonmaakverblijf	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	Onderhoudsverblijf	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	Handyman verblijf	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	EHBO / BHV / privacy ruimte/kolfruimte	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...
SRON-Leiden	Douche incl. kleedruimte	https://briefbuilder.relaticsonline.com/124cd326-e66f-4c09-bc90-f11aff00668b/ShowObject.as...



Verification

[Help](#)[Contact](#)

Project Setup > Import > Object Linker > Query Builder > **Verification** > Information Takeoff

Project:

Source RVT:

Source BB:

Document Set:

DocumentInfo RVT:

DocumentInfo BB:

Name	Description	Last Run
BB-RVT - Linked Ruimte	Presents all Briefbuilder Ruimtes that are linked to Revit Rooms	
BB-RVT - Orphaned Ruimte	Presents all Briefbuilder Ruimtes that are not linked to Revit Rooms	
BB-RVT - Orphaned Room	Presents all Revit Rooms that are not linked to Briefbuilder Ruimtes	
BB-RVT - Nr of Room	Verifies the minimum required number of Rooms in Briefbuilder can be matched t the nr of R...	
BB-RVT - Room Area	Verifies the minimum required area of Rooms in Briefbuilder is matched to the Rom Area in R...	



BriefbuilderDocument	BriefbuilderRuimte	BriefbuilderURL	RevitMass	RevitID
SRON-Leiden	Rijwielberging inpadig (180 fietsen + 8 motoren)	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	Rijwielberging inpadig	1899479
SRON-Leiden	Schoonmaakverblijf	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	Schoonmaakverblijf	2438568
SRON-Leiden	Onderhoudsverblijf	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	Onderhoudsbedrijf	2438566
SRON-Leiden	Handyman verblijf	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	Handyman verblijf	2438563
SRON-Leiden	EHBO / BHV / privacy ruimte/kolfruimte	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	EHBO/BHV/pricacy ruimte/kolfruimte	1899447
SRON-Leiden	Douche incl. kleedruimte	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	Douche incl. kleedruimten	2992465
SRON-Leiden	Werkplaats en opslag handymannen	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	Werkplaats en opslag handymannen	2412724
SRON-Leiden	Werkplaats en opslag gebouwonderhoud (externe partij)	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	Werkplaats en opslag gebouwonderhoud (externe partij)	2440139
SRON-Leiden	Werkkasten tbv schoonmaak	https://briefbuilder.relaticsonline.com/124cd326-e66f-4...	WK	2457367



Information Takeoff

[Help](#)[Contact](#)[Project Setup](#)[Import](#)[Object Linker](#)[Query Builder](#)[Verification](#)[Information Takeoff](#)

Project:	SRON
Document Set:	VO+
Source:	Revit
DocumentInfo:	SRON_EHA_1492_R18_CF_martijnAC2ZG.rvt

Object Category:	Mass
Object Property:	Gross Floor Area
Property Value:	All



No data to display

[Show In Viewer](#)

Excel export

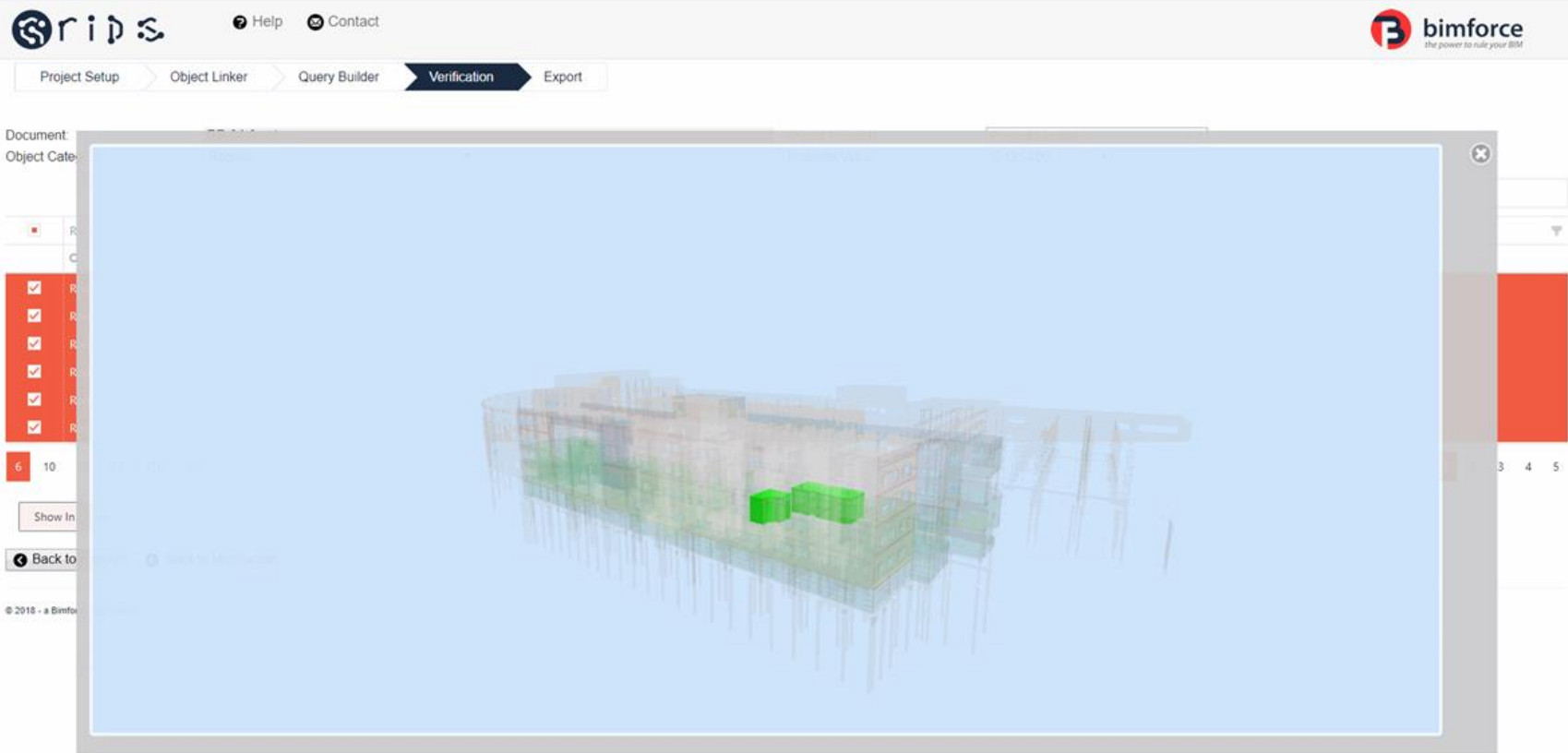
Excel spreadsheet showing data for 'Bimforce_Export.xlsx' with columns: RoomName, RoomNumber, Projectnummer, and IfcGuid. The data is organized into rows, with the first row highlighted in orange. The spreadsheet includes a ribbon with tabs: Bestand, Start, Invooien, Tekenen, Pagina-indeling, Formules, Gegevens, Controleren, Beeld, Ontwikkelaars, Help, Foxit Reader PDF, Team, and Ontwerpen. The 'Ontwerpen' tab is active, showing options for 'Verstel wat u wilt doen' and 'Tabelstijlen'.

RoomName	RoomNumber	Projectnummer	IfcGuid
Room	KA.02.013.0	C.1234.02	1NGyKexCf4W0xnljpZEYAd
Room	KA.02.013.1	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.014.0	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.014.1	C.1234.02	1NGyKexCf4W0xnljpZEYAb
Room	KA.02.014.2	C.1234.02	1NGyKexCf4W0xnljpZEYAg
Room	KA.02.014.3	C.1234.02	1NGyKexCf4W0xnljpZEYAh
Room	KA.02.014.4	C.1234.02	1NGyKexCf4W0xnljpZEYAA
Room	KA.02.014.5	C.1234.02	1NGyKexCf4W0xnljpZEYAf
Room	KA.02.015.0	C.1234.02	1NGyKexCf4W0xnljpZEYAg
Room	KA.02.015.1	C.1234.02	1NGyKexCf4W0xnljpZEYAh
Room	KA.02.015.2	C.1234.02	1NGyKexCf4W0xnljpZEYAa
Room	KA.02.015.3	C.1234.02	1NGyKexCf4W0xnljpZEYAb
Room	KA.02.015.4	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.016.0	C.1234.02	1NGyKexCf4W0xnljpZEYAd
Room	KA.02.016.1	C.1234.02	1NGyKexCf4W0xnljpZEYAd
Room	KA.02.016.2	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.029.0	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.029.1	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.030.0	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.030.1	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.030.2	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.030.3	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.030.4	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.031.0	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.031.1	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.031.2	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.031.3	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	KA.02.031.?	C.1234.02	1NGyKexCf4W0xnljpZEYAc
Room	Overig	C.1234.02	1NGyKexCf4W0xnljpZEYAc

CURRENT DEVELOPMENTS

- Integration of an IFC Viewer
- XML and XLSX as a data source
- Development of a standard set of queries and verifications
- User Management and User History implementation
- Linking GRiDS with VerteX → generating content from requirements

Integrating an IFC Viewer



User Management & History

Support Knowledge Base Contact



Personal information History Versioning Rights Company Rights User Rights

First Name 01
Middle Name 01
Last Name 01
Date of Birth Year 1974
Company 01
Department Software Engineering
Job Title MJA
Discipline HMI what?
Preferred Software C#, Neo4J
Email roel-rq2@bimforce.com
New Email (in process) roel-rq4@bimforce.com (Not Acknowledged yet)
User Name 01

Support Knowledge Base Contact

Personal information History Versioning Rights Company Rights User Rights

Object Summary

Object Name	Count
bimforce	

Generated Objects

Object	Generated On
B_conceptuele yf	2019-04-05 (10:11:06)

Object Details

Description	Value
Object Type	Walls
Manufacturer	Knauf
Language	Dutch
Information Standard	NLRS
Export Format	IFC
File Format	2x3

Property Filter Details

Description	Operator	Value
Width	>=	100
Description	like	blablabla

Support Knowledge Base Contact



Personal information History Versioning Rights Company Rights User Rights

Available Products

Vertex		
BIDap		
Print		

Save

Removal of already existing products is not possible!

c33b62a-b20f-481d-b0db-
9c4796f69d30

Interfaces

No interfaces available yet		
Serial UI for Vertex		
Vertex Revit Generator		

Save

Removal of already existing interfaces is not possible!

61f69b3a-1994-4bc7-a155-
66ea64b1b3c8

Interface Rights

Name	Description	Value

Save

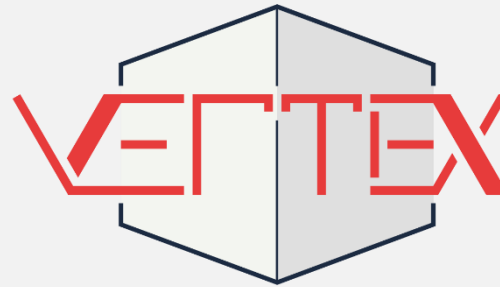
Removal of already existing rights is not possible!

TEST

2019 Bimforce - All rights reserved
Current User: ref00710-b0a2-4608-42d2-ec080147f920

VerteX & GRiDS

- VerteX is a data-driven Content Management Platform
- Unlike other Content solutions it's completely data-driven
- Which allows us to connect it to GRiDS & re-use data from other sources in your 3D model



VerteX & GRiDS



[Support](#) [Knowledge Base](#) [Contact](#)

Object Filtering

Select Manufacturer:

UMC

Select Object Category:

Room

Property Filtering

property	operator	value
-- Select property (11)	cont	
-- Select property (11)		
-- Select property (11)		
-- Select property (11)		
-- Select property (11)		

Show Result

Reset Filters

results (2566) ^ v

WKZ KB 01 056 4

WKZ KB 01 041 1

WKZ KB 01 041 0

WKZ KB 01 023 5

WKZ KB 01 057 5

WKZ KB 01 057 4

WKZ KB 01 056 5

WKZ KB 01 023 6

WKZ KB 01 022 1

WKZ KB 01 044 0

WKZ KB 01 056 6

WKZ KB 01 055 0

WKZ KB 01 042 1

Generate Object

Properties (Single item only)

Preview_direct

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Property	Value
Bruto vloeroppervlak	15.5 m2
Code	WKZ KB.01.057.4
Code bouwdeel	B
Code bouwlaag	01
Code gebouw	WKZ
Code kostenplaats	I100
Manufacturer	UMC
Netto vloer oppervlak	16.06 m2
Omschrijving	Familiekamer
Omschrijving functiesoort	ALG.VERPLEGING
Omschrijving omgeving	FACILITAIREOMG
Ruimtegebruik	Familiekamer
Ruimtesoort	Persoonsruimte
Tarra vloer oppervlak	0 m2
Type Name	WKZ KB.01.057.4



VerteX & GRiDS



SupportKnowledge BaseContact

Object Filtering

Select Manufacturer:SRON

Select Object Category:-- Select --

Property Filtering

property	operator	value
-- Select property (40)		
-- Select property (40)		
-- Select property (40)		
-- Select property (40)		
-- Select property (40)		

Show ResultReset Filters

results (590)

VID0450 - LPKF printfreesmachine_

VID0451 - LPKF printfreesmachine_Werktafel

VID0452 - LPKF printfreesmachine_PC

VID0453 - Werkbank

VID0454 - Werkbank_Printboormachine

VID0455 - Werkbank_metaalknipper

VID0456 - Werkbank_Lintzaag

VID0457 - Werkbank_Bankschroeven

VID0458 - Werkbank_buigknijptool

VID0459 - Werkbank_Soldeerbout

VID0460 - Ruimte voorzieningen_

VID0461 - Chemiekast

VID0462 - Belichtingsbak

Generate Object

Properties (Single item only)

Preview_direct

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powered by GeometryGym



Property	Value
Category	Generic Model
Heatload (berekening controleren!) Afzuiging	0 kW
Heatload (berekening controleren!) Koelwater	0 kW
Heatload (berekening controleren!) Omgeving Gelijktijdig	0 kW
Heatload (berekening controleren!) Omgeving Omgeving	0 kW
Heatload (berekening controleren!) Totaal	0 kW
Locatie Huidig	S.065
Locatie PVE SRON	4.1.14a
Manufacturer	SRON
Omschrijving geheel	Werkbank
Onderdeel	buigknijptool
Type Name	VID0458 - Werkbank_buigknijptool
Verhuis-ID	VID0458



FUTURE DEVELOPMENTS

- More connections:
 - Document Management Systems
 - DWG
 - Maintenance software
 - Security / Building Controls
 - Logistics
 - GIS

LESSONS LEARNED

Lessons Learned

- Plan extensively. Preperation is key!
Don't expect your project to go according to plan. Try, fail, try harder
- Strategic changes to business processes should be modular implementations in order to deal with scope changes.
- The key aspect for BIM in FM is managing the aspect of time. Time changes everything:
 - Data is locked in (soon to be outdated) software and/or files
 - Project workflows change from linear to diffuse
- There are no readily available solutions for BIM in FM that deal with these aspects.

REMINDER:

Speaker Feedback is appreciated

Fill in the Survey on the Mobile App





S2.5 Case Study - Developing a new FM strategy for 610.000m2 real estate

Martijn de Riet

Bimforce

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